BookWorld

Software Requirements Specification

Version 1.0

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Revision History

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1. Introduction

1.1 User Story

- Prof. Punit Patel is managing a Library manually which consists of 750 well-known life changing books. Managing all the Book entries manually, tracking each issued book, answering queries of the issuer, conflicting book demands, timely return of the issued books and keeping track of the donors who donated books for a good cause is sometimes a task of concern.

- Prof. Punit Patel was facing this problem, so he approached us for the solution. As per his requirements we are developing this software named BookWorld.

1.2 Purpose

- The main purpose is to customize a Software that helps Mr. Punit Patel to reduce his workload and helps him to manage things smoothly. The Software aims at reducing manual labor.

1.3 Scope

- BookWorld provides a software interface with features of Managing Books, tracking issued books and the deadline to return it, tracking users and few other features as per given below:
- Login for Admin
- Book Registration by Admin
- Track record of users by Admin
- System generated Notification as due date approaches to return the book by Admin
- Login/Register for User
- Book Issue by User
- Donate book by User
- Checking for Book Availability by User
- Categorical Filtration based on the Theme, Author by the User
- Contact Us tab for User

1.4 Overview

- The SRS will provide a detailed description of BookWorld which is a Library Management System. This document will provide the outline of the requirements, overview of the characteristics and constraints of the system.

- **Section 2:** This section of the SRS will provide the general factors that affect the product and its requirements. It provides the background for those requirements. The items such as product perspective, product function, user characteristics, constraints, assumption and dependencies are described in this section.

- **Section 3:** This section of SRS contains all the software requirements mentioned in section 2 in detail, sufficient enough to enable the Designers to design the system to satisfy the requirements and Testers to test if the system satisfies those requirements.

2. Overall Description

2.1 Product Perspective

The BookWorld is a software to be used by our client Prof. Punit Patel to digitalize the Library Management System which he has been maintaining manually. The development of Bookworld will be beneficial for our Client as well as the Users. The BookWorld project aims to develop a comprehensive book management platform to assist Prof. Punit Patel to efficiently manage his book inventory, book issuing process, and ensure timely notifications to the users when book duration is about to expire.

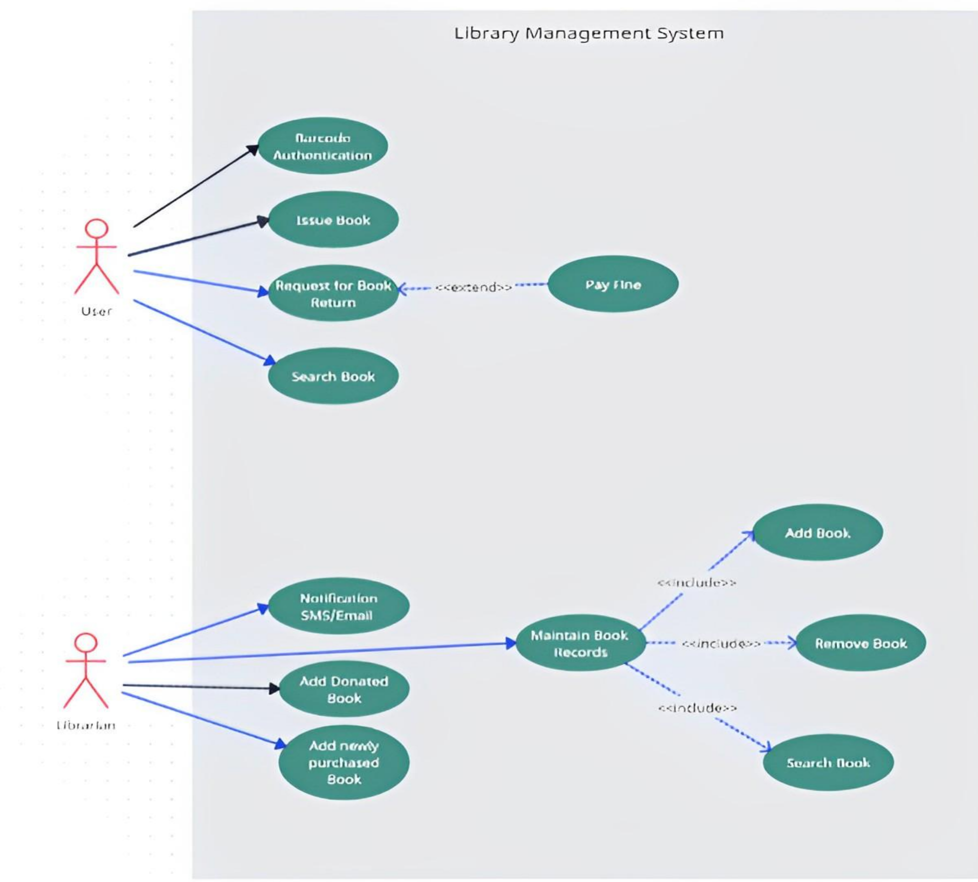

Figure 1 : UseCase Diagram

2.2 Product Functions

The functions of the system includes :

1. Login for Admin
2. Book Registration by Admin
3. Track record of users by Admin
4. System generated Notification as due date approaches to return the book by Admin
5. Login/Register for User
6. Book Issue by User
7. Donate book by User
8. Checking for Book Availability by User
9. Categorical Filtration based on the Theme, Author by the User
10. Contact Us tab for User

2.3 User Characteristics

The users of the system can be the students of the college or any faculty member. The administrator of the system is assumed to have basic knowledge of computer. The appropriate user interface, user's manual, online help and the guide to install and maintain the system must be efficient to educate the admin on how to use the system without any problem. The users can access the system easily through it's user-friendly interface.

2.4 Constraints

1. The Information regarding all of the books should be stored in the database that is accessible by BookWorld.
2. BookWorld should be installed on the Administrator's computers.
3. The users can access BookWorld from any device that has an Internet connection.
4. The users must have their correct username and passwords to gain access to their accounts on BookWorld.

2.5 Assumptions and Dependencies

1. The Administrator should have basic knowledge of the computer.
2. Admin's computer should have an Internet Connection.

3. Users accessing the website should have an Internet Connection.

3. Specific Requirements

- This Section describes all the Functional as well as the Non-functional requirements. It contains description of functions and capabilities that the product must provide.

3.1 Functional Requirement

- Functional requirements describe how a product must behave, what are its features and functionalities.

F1 : Login for Admin

The admin can log in to the system using their username and password.

- Input : Admin Username and Password.
- Output : Admin Dashboard with access to administrative features.
- Processing : The system verifies the admin's credentials against the database, and if valid, grants access to the Admin Dashboard.

F2 : Book Registration by Admin

The admin can register new books into the system with relevant details.

- Input : Book information, including title, author, ISBN, genre, and other relevant data.
- Output : Newly registered book added to the library catalog.

F3 : Track record of Users by Admin

The admin can view and track the borrowing history and details of library users.

- Output : User details and their borrowing history displayed to the admin.

F4 : System generated Notification as due date approaches to return the book by Admin

The system automatically generates notifications for users as their book due dates approach.

- Output : Notifications sent to users with details of overdue books.

F5 : Authentication and Authorization for User

- Login : The user can login to the system with his/her username and password.
    - Input : Username and Password.
    - Output : User Dashboard , Admin Dashboard
    - Processing : Username and password is verified from the database and if a user exists in the database then the user interface will be displayed according to their role.
- Logout : The user can logout from the system.
    - Input : 'Logout' option is selected.
    - Output : User login Screen will be displayed.
- Password reset : Users can reset their password..
    - Input : Username/email
    - Output : Reset password page is provided to the user
    - Processing : If a user exists in the database then a mail will be sent to the user through which user can reset their passwords.

F6 : Book Issue by User

Registered users can request to borrow books from the library.

- Input : Book details for the book they want to borrow.
- Output : Confirmation of book issue and due date provided to the user.

F7 : Donate book by User

Users can donate books to the library.

- Input : Book information, including title, author, ISBN, genre, and any other relevant data.
- Output : Confirmation of successful book donation.

F8 : Checking for Book Availability by User

Users can check the availability status of books in the library.

- Input : Book details or ISBN for the book they want to check.
- Output : Availability status displayed to the user.

F9 : Categorical Filtration based on the Theme, Author by the User

Users can filter and search for books based on categories such as theme or author.

- Input : User's chosen filter criteria.
- Output : List of books matching the selected filter criteria.

F10 : Contact Us tab for User

Users can access a "Contact Us" feature to get in touch with the library staff.

- Output : User can access contact information or a form to send inquiries or feedback.

3.2 Non-Functional Requirement

3.2.1 Usability

- Usability defines how difficult it will be for a user to learn and operate the system.

Efficiency of use:

- User can easily interact with system. Most of the tasks, a user can complete without any help. It does not have a complex design as a result, any user can easily interact with the system.

3.2.2 Reliability

- Reliability defines how likely it is for the software to work without any failure for a given period of time. Reliability decreases because of bugs present in the code, hardware failures, or problems with other system components.

- The Database update process must roll back all related updates when any update fails.

3.2.3 Performance

- Performance is a quality attribute that describes the responsiveness of the system to various user interactions with it.

- The front-page load time must not be more than 5 seconds.

3.2.4 Security

- Security requirements ensure that the software is protected from unauthorized access to the system and its stored data. It considers different levels of authorization and authentication across different *users'* roles. For instance, data privacy is a security characteristic that describes who can create, see, copy, change, or delete information.

- Unauthorized user cannot login to system.

3.3 Design Constraints

3.3.1 No cost for cloud :

- Client has denied to pay monthly for cloud, so we have to design accordingly.

3.3.2 Software must not have any dependency and must be lightweight :

- Software must not have any dependency so can be used in any system.

- It must be light weight so employee don't have to wait much or face any difficulty in their work.